

Problem-Solution Fit

Date	29 September 2026
Team ID	SWTID-2026-7019
Project Name	EduGenie - AI Learning Assistant
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] - School and high-school students (Grades 6-12) who study on their own after class hours. - College freshers and self-learners picking up new skills (SQL, Python, ML). - Teachers/tutors needing quick explanations and quiz material.	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Limited budget - cannot afford private tuition or paid subscriptions. - Basic phones/laptops and uneven internet connectivity. - Little time; low patience for long or jargon-heavy content. - No expert on hand to check whether an answer is right.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Search/YouTube: free, huge; but unstructured, ads, no personalisation. E-learning apps: structured; but paid, fixed syllabus, no free-form doubts. General chatbots: flexible; not student-focused, one tool per need. Private tutors: personal; costly, not on demand.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - No instant, simple answer to a doubt - daily. - Chapters/notes too long to digest - several times a week. - No easy way to self-test after studying - weekly. - Unsure what to learn next when starting a topic - each new topic.	9. PROBLEM ROOT / CAUSE - Doubts arise outside class or tutor hours. - Study material is written for experts, not for learners. - Doubt-solving, summaries, quizzes and roadmaps live in separate tools. - Making practice quizzes by hand is slow. - No guided, level-wise learning path.	7. BEHAVIOR + ITS INTENSITY [BE] - Googles every doubt and opens several videos/pages (high: 30-60 min per doubt). - Copy-pastes text between chatbots and notes (medium). - Skips self-testing before exams (high). - Drops a topic when it feels overwhelming (medium).
3. TRIGGERS TO ACT [TR] - Homework or assignment doubt. - Upcoming test or exam. - A long or difficult chapter to revise. - Wanting to start a new skill or subject. - Teacher or friend recommends a topic.	10. YOUR SOLUTION [SL] EduGenie - one web-based AI learning assistant, powered by Google Gemini, with a local Flan-T5 model as fallback for explanations. Ask a Question: instant answers to any doubt. Explain: concepts in simple, school-level language. Summarize: turns long passages into short, plain-language notes. Quiz: auto-generates 3 multiple-choice questions from any text, with instant answer feedback. Learning Path: beginner-to-advanced roadmap with topics, order and resources. - All five tools sit on a single page, respond in real time and cost nothing for the learner to use.	8. CHANNELS OF BEHAVIOR [CH] ONLINE - Google Search, YouTube, e-learning sites, general AI chatbots. - WhatsApp/Telegram study groups. EduGenie web app in any browser (FastAPI backend).
4. EMOTIONS BEFORE / AFTER [EM] Before: confused, anxious, overwhelmed by information, low confidence. After: clear, confident, motivated and in control of their own learning.		OFFLINE - Classroom and teacher doubt sessions. - Textbooks, handwritten notes, library. - Private tuition and peer study groups.